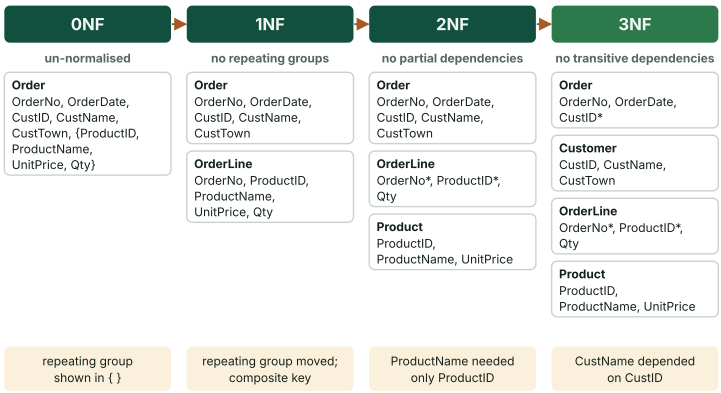
Normalisation to third normal form

IS03 FACT FILE

Specification: students should be able to

- Explain the purpose of normalisation
- Normalise data from un-normalised form (0NF) to first, second and third normal form
- Identify partial and transitive dependencies

Key words: normalisation, 0NF, first normal form, second normal form, third normal form, repeating group, atomic, functional dependency, partial dependency, transitive dependency, update anomaly, insertion anomaly



Fermanagh Fresh order data normalised from 0NF through 1NF and 2NF to 3NF, showing which dependency is removed at each stage

Purpose of normalisation

Normalisation is a step-by-step process for structuring data into well-designed relations. It minimises redundancy and so prevents update, insertion and deletion anomalies. An update anomaly occurs when the same fact stored in several rows is changed in some but not all of them. An insertion anomaly is being unable to add data because other unrelated data is missing. A deletion anomaly is losing useful data when another record is removed.

The three normal forms

First normal form removes repeating groups so each field holds a single atomic value; the repeating group goes into a new table carrying the original key, which becomes part of a composite key. Second normal form removes partial dependencies, where a non-key attribute depends on only part of a composite key. Third normal form removes transitive dependencies, where a non-key attribute depends on another non-key attribute. A relation with a single-attribute key in 1NF is automatically in 2NF.

Key terms from the mark schemes

Normalisation
A formal process of organising data into relations to minimise redundancy and prevent update, insertion and deletion anomalies.

First normal form (1NF)
A relation with no repeating groups in which every attribute holds a single atomic value.

Second normal form (2NF)
A relation in 1NF with no partial dependencies on a composite key.

Third normal form (3NF)
A relation in 2NF with no transitive dependencies, so every non-key attribute depends only on the primary key.

Partial dependency
Where a non-key attribute depends on only part of a composite primary key.

Transitive dependency
Where a non-key attribute depends on another non-key attribute rather than directly on the primary key.

Update anomaly
An inconsistency caused when a fact stored in several places is changed in some but not all of them.

Exam focus

Fermanagh Fresh order forms in 0NF become Customer(CustID, CustName, CustTown), Order(OrderNo, OrderDate, CustID*), Product(ProductID, ProductName, UnitPrice) and OrderLine(OrderNo*, ProductID*, Qty). ProductName was partially dependent on ProductID alone; CustName was transitively dependent on CustID. Each fact is now held once, and the four tables can be rejoined using SQL whenever a full order is needed.

? Check yourself

1. State two reasons why data is normalised. [2]
2. Define a partial dependency. [2]
3. Define a transitive dependency and give an example. [2]

Answers: 1. Any two from: to minimise / reduce data redundancy (1) / to prevent update anomalies and so data inconsistency (1) / to prevent insertion anomalies (1) / to prevent deletion anomalies (1) / to make the database easier to maintain and change (1). 2. A partial dependency exists where a non-key attribute depends on only part of a composite primary key (1) / rather than on the whole key, e.g. in OrderLine(OrderNo, ProductID, ProductName, Qty) ProductName depends only on ProductID (1). 3. A transitive dependency is where a non-key attribute depends on another non-key attribute rather than directly on the primary key (1) / e.g. in Order(OrderNo, OrderDate, CustID, CustName), CustName depends on CustID, which is not the key (1). Accept any correct example, e.g. TutorName depends on TutorID.